Vincenzo Marcianò *AI Research Scientist and Engineer*

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Education

Visiting Researcher, *King's College London* – London, United Kingdom — Sep 2023 - Sep 2026
PhD in AI & Computer Science, *Sorbonne University* – Biot, France — Sep 2023 - Sep 2026
MSc in Data Science & Engineering, *Polytechnic University of Turin* – Turin, Italy. GPA: 3.8/4.0 — Oct 2020 - Dec 2022
Exchange program for Msc Thesis, *The University of Sheffield* – Sheffield, United Kingdom — Feb 2022 - Jul 2022
BSc in Electronics Engineering, *Polytechnic University of Turin* – Turin, Italy. GPA: 3.0/4.0 — Oct 2017 - Jul 2020

Experience

Applied Scientist Intern, *Amazon (Alexa)* – Turin, Italy — Jun 2026 – Ongoing

- Developing benchmarks for stress-testing non-stationarity in agentic LLM routing.
- Designed a multi-arm-bandit router leading to a **10%** decrease in API calls and **10%** decrease in regret.
- Submitted a paper to **ICLR '27** achieving a **30%** in performance decrease for state-of-the-art LLM routing algorithms under custom adversarial injection.

Data and Research Scientist, *LexSA* – Milan, Italy — Jan 2026 – May 2026

- Built an ad-hoc web-scraper that rapidly navigates across the Fiscal Profile of each user and provides fiscal details processed by an LLM. Scraped and embedded more than **1.2 Million** juridic public documents.
- Back-end development of custom agentic skills for online fiscal tools. Improved retrieval accuracy of **50%**.

Doctoral Researcher, *EURECOM (AI4Health Lab)* – Biot, France — Sep 2023 – May 2026

- Designed and engineered new state-of-the-art refinement and quality control frameworks for image segmentation, multimodal diffusion models and foundation-model-driven visual prompt sampling. Achieved SOTA performance with an average **15%** increase.
- Involved in agentic AI projects for automated medical segmentation pipelines, and contributed to three open-source multimodal medical imaging frameworks.

Applied Scientist Intern, *Amazon* – Barcelona, Spain — Sep 2022 – Mar 2023

- Reduced forecasting error (WAPE) by **30%** by designing time-domain embeddings for multivariate time-series forecasting, enhancing accuracy and stability across vendor datasets.
- Improved model scalability and robustness by generalizing forecasting models across heterogeneous data granularities used in large-scale supply chain operations.

Publications and Relevant Projects

- **Marcianò, V.**, et al. *Diffusion-Based Quality Control of Medical Image Segmentations across Organs*. IEEE TMI.
- **Marcianò, V.**, et al. *Retrieval-Augmented Correction for Generalist Medical Image Segmentation*. Under review, TMLR.
- **Marcianò, V.**, et al. *Mask-Image Distributional Divergence for Evaluating Image Segmentation*. Under review (Provisional Accept), NeurIPS.
- Poeta, E., Vargas, L., Falcetta, D., **Marcianó, V.** et al. (2025) *Divergence-Aware Training with Automatic Subgroup Mitigation for Breast Tumor Segmentation*. In: Gee, J.C., et al. MICCAI 2025 Workshops. **[Best Paper Award]**
- Chaptoukaev, H., **Marcianó, V.**, Galati, F., Zuluaga, M.A. (2024). *HyperMM: Robust Multimodal Learning with Varying-Sized Inputs* MICCAI 2024 Workshops. **[Best Paper Award]**
- **Overcomplete (Research Contribution, Harvard Lab):** Contributed to the official implementation of RA-SAE.

Skills

Programming & Core: Python, C++, C, Bash
ML & Medical Imaging: PyTorch, HuggingFace, vLLM, Tensorflow, MONAI, TorchIO, scikit-learn, scikit-time, ITK-SNAP
Model Architectures: Diffusion Models (DDIM, FM, DiT), VAEs, VLMs, LLMs, RAG, Graph-RAG, Bayesian Routers
Infrastructure & Tools: HPC Clusters, Docker, CUDA, Git/GitHub, Linux/Unix, LaTeX
Languages: Italian (Native), English (Advanced), French (Intermediate), Spanish (Basic)